

RESEARCH ARTICLE

Aging and Adaptation to Exercise

Higher resistance training volume offsets muscle hypertrophy nonresponsiveness in older individuals

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Abstract

The magnitude of muscle hypertrophy in response to resistance training (RT) is highly variable between individuals (response heterogeneity). Manipulations in RT variables may modulate RT-related response heterogeneity; yet, this remains to be determined. Using a within-subject unilateral design, we aimed to investigate the effects of RT volume manipulation on whole muscle hypertrophy [quadriceps muscle cross-sectional area (qCSA)] among nonresponders and responders to a low RT dose (single-set). We also investigated the effects of RT volume manipulation on muscle strength in these responsiveness groups. Eighty-five older individuals [41M/44F, age = 68 ± 4 yr; body mass index (BMI) = 26.4 ± 3.7 kg/m²] had one leg randomly allocated to a single (1)-set and the contralateral leg allocated to four sets of unilateral knee-extension RT at 8–15 repetition maximum (RM) for 10-wk 2 days/wk. Pre- and postintervention, participants underwent magnetic resonance imaging (MRI) and unilateral knee-extension 1-RM strength testing. MRI typical error (2 × TE = 3.27%) was used to classify individuals according to responsiveness patterns. *n* = 51 were classified as nonresponders (≤2 × TE) and *n* = 34 as responders (>2 × TE) based on pre- to postintervention change qCSA following the single-set RT protocol. Nonresponders to single-set training showed a dose response, with significant time × set interactions for qCSA and 1-RM strength, indicating greater gains in response to the higher volume prescription (time × set: *P* < 0.05 for both outcomes). Responders improved qCSA (time: *P* < 0.001), with a tendency toward higher benefit from the four sets RT protocol (time × set: *P* = 0.08); on the other hand, 1-RM increased similarly irrespectively of RT volume prescription (time × set: *P* > 0.05). Our findings support the use of higher RT volume to mitigate nonresponsiveness among older adults.

NEW & NOTEWORTHY Using a within-subject unilateral design, we demonstrated that increasing resistance training (RT) volume may be a simple, effective strategy to improve muscle hypertrophy and strength gains among older adults who do not respond to low-volume RT. In addition, it could most likely be used to further improve hypertrophic outcomes in responders.

aging; muscle mass; muscle strength; resistance exercise; sets

INTRODUCTION

Resistance training (RT) is a well-known intervention to mitigate the onset of age-related declines in skeletal muscle mass and function (1–4). Nonetheless, not all individuals benefit from a standardized RT intervention to a similar degree. For instance, while RT confers a range of physiological benefits, the magnitude of RT-induced muscle hypertrophy is highly variable between individuals, ranging from minor (i.e., low- or nonresponders) to robust gains (i.e., high-responders or responders) (5–11). Despite this, it has been suggested that adjustments in exercise stimulus could mitigate nonresponsiveness across different exercise types and outcomes (12–15). In fact, previous data indicate that manipulations of both weekly volume and intensity during RT yielded a lower nonresponse rate (11). Damas et al. (16) showed that the only training variable manipulation that augmented acute myofibrillar protein synthesis was the increase in RT volume (number of sets within an RT session), with no effect of several other variables (i.e., intensity, pause, and muscle action). In addition, higher exercise volume has been shown to increase activation of p70S6K (17, 18)—a downstream target of mammalian target of rapamycin (mTOR), suggesting a potential role for RT volume on responsiveness.

Notwithstanding the aforementioned, literature is still equivocal on whether increasing exercise volume affects variability in hypertrophic responsiveness to RT. Apart from divergencies between exercise protocols, sample characteristics, and duration of the intervention (5–11), other intrinsic mechanisms (6, 9, 10, 19–22) and environmental factors (e.g., sleep, nutrition, and physical activity levels) thought to be linked with response heterogeneity (present in between-subject study designs) may also contribute to this phenomenon. In this context, within-subject experimental designs may help mitigate some of the between-subject variability and provide more precise insight into the effects of RT volume manipulation at the individual level.

RT also promotes functional adaptations (e.g., increases in muscle strength). Not surprisingly, the magnitude of these responses is also highly variable across individuals (5, 23, 24). Considering the hypothesis that RT volume manipulation could improve nonresponsiveness, potentiating the hypertrophy at whole muscle tissue, one could also expect improvements at the functional level. In fact, retrospective data suggest that individuals can benefit from RT interventions exhibiting at least one adaptation in response to training (23), though only a fraction shows improvements in all RT-related outcomes.

Thus, the present study aimed to investigate the effects of RT volume manipulation on whole muscle tissue hypertrophic adaptations among nonresponders and responders to a low RT dose (single-set). A secondary purpose was to investigate the effects of RT volume manipulation on muscle strength in these responsiveness groups. We hypothesized that increases in RT volume would mitigate nonresponsiveness and would enhance the hypertrophic response even further on responders. We also hypothesized that increases in RT volume would result in similar effects for muscle strength.

METHODS

Participants

Eighty-five healthy older individuals (>60 yr of age) of both sexes participated in the study (Table 1). Exclusion criteria included diagnosed type I diabetes, ischemic myocardial disease, arrhythmia, noncontrolled hypertension, any major orthopedic musculoskeletal disorder, and use of any dietary supplement or drug with potential anabolic effects. Participants were not engaged in any kind of regular resistance training and/or aerobic training for at least six mo before the experimental period. Participants provided their written informed consent. The experimental protocol was approved by the Human Research Ethics Committee of the University of Sao Paulo.

Experimental Design

Before the 10-wk RT intervention period, bilateral quadriceps muscle cross-sectional areas (qCSA) were assessed via magnetic resonance imaging (MRI). Then, participants performed one familiarization session with the unilateral knee-extension one-repetition maximum test (1-RM). Seventy-two hours after the familiarization session, participants performed the unilateral 1-RM tests. All participants underwent a baseline nutritional evaluation by a trained dietitian. After that, each participant had one leg randomly allocated in counter-balanced fashion to single (1)-set (2 weekly total) and the contralateral leg was allocated to four sets (8 weekly total) of unilateral knee-extension resistance exercises at 8–15 RM for 10 wk of unilateral RT 2×/wk. After the intervention, participants underwent an MRI examination by the 1-RM test for each leg.

To determine the effects of RT volume manipulation on whole tissue hypertrophy at the individual level, we used the MRI typical error ($2 \times \text{TE} = 3.27\%$) to allocate individuals according to their responsiveness patterns (refer to *Quadriceps Muscle Cross-Sectional Area* for additional details). Thus, we classified individuals as nonresponders

Table 1. Participants' characteristics and nutritional intake at baseline

Variables	Nonresponders (n = 51)	Responders (n = 34)	P
Age, yr	69 ± 5	68 ± 4	0.40
Body mass, kg	72.2 ± 12.9	69.7 ± 11.5	0.35
BMI, kg/m ²	26.4 ± 3.9	26.1 ± 3.5	0.69
Sex, n male/female	27/24	14/20	—
Nutritional intake			
Energy, kcal/day	2,105.3 ± 590	1,916.9 ± 617.1	0.18
Protein, g/kg/day	1.6 ± 0.3	1.5 ± 0.4	0.23
Protein (% TEI)	22.2 ± 4.2	22 ± 3.6	0.76
Carbohydrates (% TEI)	47.5 ± 7.2	50.4 ± 6.7	0.07
Fat (% TEI)	30.2 ± 6.6	27.6 ± 5.6	0.06

Data are presented as means ± standard deviation. Responsiveness groups based on pre- to postintervention change (%) for quadriceps cross-sectional area (qCSA) to 1 set. Intake values include two 20-g doses/day of whey protein concentrate. 1 set, single-set unilateral resistance exercise; BMI: body mass index; nonresponders, qCSA change $\leq 2 \times$ typical error for magnetic resonance imaging ($2 \times \text{TE} = 3.27\%$) for 1 set; responders, qCSA change $> 2 \times \text{TE}$ for 1 set; TEI: total energy intake.

($\leq 2 \times \text{TE}$) or responders ($> 2 \times \text{TE}$) (25–32), based on the qCSA pre- to postintervention change (%) for the lower RT volume (1 set).

Quadriceps Muscle Cross-Sectional Area

Quadriceps muscle CSA was obtained through magnetic resonance imaging (MRI) (Sigma LX 9.1, GE Healthcare, Milwaukee, WI). Participants lay on the device in a supine position and had their knees extended and tied with velcro straps (33). Images were acquired at 50% of the segment length—between the perpendicular distance from the greater trochanter to the inferior border of the lateral epicondyle of the femur. The qCSA was measured by a blinded researcher to the RT volume allocation using ImageJ software (v.1.53c, National Institutes of Health, Bethesda, MD) as previously described by our group (34).

The reproducibility for MRI was calculated on a subsample of 20 participants enrolled in the study. Before the intervention, these participants performed two MRI examinations interspaced by 7 days. The MRI typical error (TE) was calculated accordingly: $\text{TE} = \text{standard deviation of the difference between two repeated measures interspaced by 7 days} (\text{day } 2 - \text{day } 1) \times \sqrt{2}^{-1}$ (26). The TE expressed as a coefficient of variation was 1.63%. The MRI measurement error was defined as 2 times the TE ($2 \times \text{TE} = 3.27\%$).

Maximum Dynamic Strength (1-RM)

Unilateral quadriceps 1-RM of both legs were assessed on a knee-extension machine following previous recommendations (35). At the first visit, participants were familiarized with the 1-RM test. The protocol consisted of a 5-min general warm-up on a cycle ergometer. This was followed by a specific warm-up of one set of eight repetitions at 50% of the estimated 1-RM, followed by one set of three repetitions at 70% of the estimated 1-RM with a 1 min rest between sets. After 3 min of rest, participants had up to five attempts to achieve their 1-RM. The smallest incremental load for the next attempt was ~ 1 kg. A 3-min rest interval was allotted between attempts and the higher load achieved (full eccentric-concentric movement with 90° range of motion) was considered 1-RM. After 48 h, participants performed 1-RM test sessions interspaced by 72 h. The highest value obtained between sessions was used as the 1-RM.

The reproducibility for 1-RM was calculated based on both tests before the intervention. The TE expressed as a coefficient of variation was 4.44%. The 1-RM measurement error was defined as 2 times the TE ($2 \times \text{TE} = 8.90\%$).

Dietary Assessment

Participants performed three 24-h dietary recalls (2 non-consecutive weekdays and 1 weekend day) to determine habitual daily nutrient intake, using the USDA Automated Multiple-Pass Method (36). A trained nutritionist gave oral and written instructions on how to perform the dietary recalls, including timing and estimation of portion size using household measures on an individual basis (37, 38). Upon retrieval, the same nutritionist confirmed the accuracy of estimated portions of the dietary recalls using a visual album with real photos of foods (39). Data were analyzed with specialized nutrition software (Nutritionist Pro version 7.3, Axxya

Systems, Woodinville, WO). To mitigate any potential confounding effect of protein intake in our experimental design, all participants were provided with a high-quality protein supplement (~ 20 g of whey protein concentrate) and instructed to consume it twice a day (in the morning [at least 2–3 h before lunch] and 30 min before sleep) during the experimental period to meet daily protein intake recommendation for older adults (≥ 1.2 g/kg body wt/day) (40). Participants were instructed to maintain their normal dietary habits and not to consume any other additional supplement throughout the intervention. Due to technical issues, we missed data for the dietary assessment analysis for five participants.

Resistance Training Protocol

The RT protocol was performed using a unilateral knee extension machine. One leg performed one set whereas the contralateral performed four sets of 8–15 RM with a 60–90 s interval between sets. To reduce potential sources of bias to exercise order, participants alternated the starting leg for each training session. During the first 2 wk, participants performed all sets within the 12–15 RM range. From the third week onward, participants performed all sets at 8–12 RM. To maintain the pre-established range of repetitions, the load was adjusted through sets and RT sessions. For instance, if participants were above the target repetitions range, the load was increased for the next set; if the participants were below the target repetitions range, the load was reduced for the next set. The smallest load adjustments were 1 kg. Participants performed each set to concentric muscle failure (i.e., inability to complete the repetition within the full range of motion). All training sessions were supervised by an experienced exercise physiologist. Total training volume is expressed as the sum of the volume load in each training session (sets $\times$ repetitions $\times$ load).

Statistical Analyses

The qCSA and 1-RM were analyzed by a mixed model for repeated measures, with time (pre- vs. postintervention) and sets (1 set vs. 4 sets) as fixed factors and participants as random factors. Analyses were conducted on each responsiveness group separately. In addition, we used the pre- to postintervention absolute change for qCSA and 1-RM for the comparison between nonresponders versus responders using a mixed model for repeated measures, with responsiveness groups (nonresponders vs. responders) and sets (1 set vs. 4 sets) as fixed factors and participants as random factors.

Differences between responsiveness groups (nonresponders vs. responders) for age, body weight, body mass index, nutritional intake, and training adherence were tested with independent *t* tests. We also compared total training volume between responsiveness groups using a mixed model for repeated measures with responsiveness group (nonresponders vs. responders) and set (1 set vs. 4 sets) as fixed factors, and participants as random factors.

We determined the presence of extreme observations and the normal distribution of residuals through residual analyses. Whenever a significant *F* value was obtained, Tukey's post hoc test was used for pairwise comparisons ($P < 0.05$).

Analyses were performed using SAS 9.3 (SAS Institute Inc., Cary, NC), R statistical software (v.3.6.3, R Foundation for Statistical Computing Vienna, Austria), and Rstudio (v.1.4.1103, Rstudio, Inc., Boston, MA). All data are presented as means $\pm$ standard deviation or estimated mean difference (EMD) and the 95% confidence interval derived from the mixed model unless stated otherwise. The significance level was set at $P < 0.05$.

RESULTS

Participant Flow

A total of 186 older adults were screened to participate in the present study. Forty participants did not meet the inclusion criteria and 30 declined to participate, lost interest, or had a conflicting schedule. Fourteen participants dropped out of the study before baseline assessments, and 102 participants were randomly allocated to the intervention. Fourteen participants discontinued due to personal reasons and three participants were removed from the analysis because they reported being trained only after starting the intervention. A total of 85 participants finished the intervention and were included in the analysis (Fig. 1).

Participant Characteristics and Nutritional Intake at Baseline and Training Adherence

Participants' characteristics and nutritional intake at baseline are described in Table 1. Fifty-one participants were classified as nonresponders (60%) and 34 were classified as responders (40%). No statistically significant differences were observed between nonresponders versus responders for age, body mass, or body mass index (all $P >$

0.05). In addition, training adherence was similar between responsiveness groups (nonresponders: $97.3 \pm 4.5\%$ vs. responders: $98.5 \pm 2.6\%$; $P = 0.1$). No statistical differences were observed between nonresponders and responders for nutritional intake variables (all $P > 0.05$) (Table 1).

Quadriceps Muscle Cross-Sectional Area

Nonresponders increased the qCSA pre- to postintervention only in response to 4 sets (time $\times$ set: $P < 0.001$; EMD: 3.02 cm^2 , 95% CI: $2.06\text{--}3.99 \text{ cm}^2$, $P < 0.0001$), with no effect for 1 set (EMD: 0.49 cm^2 , 95% CI: $-0.46\text{--}1.46 \text{ cm}^2$, $P = 0.54$). The qCSA was statistically higher at postintervention for the 4 sets protocol compared with 1 set (EMD: 2.12 cm^2 ; 95% CI: $1.15\text{--}3.19 \text{ cm}^2$, $P < 0.0001$) (Table 2). In contrast, responders improved qCSA following RT intervention (time: $P < 0.001$; EMD: 3.48 cm^2 , 95% CI: $2.73\text{--}4.23 \text{ cm}^2$), but with no statistical difference between sets (time $\times$ set: $P = 0.08$; 1 set EMD: 2.81 cm^2 , 95% CI: $1.41\text{--}4.21 \text{ cm}^2$; 4 sets EMD: 4.15 cm^2 , 95% CI: $2.75\text{--}5.55 \text{ cm}^2$; 1 set vs. 4 sets postintervention EMD: 2.11 cm^2 , 95% CI: $0.72\text{--}3.51 \text{ cm}^2$) (Table 2). Comparison between nonresponders versus responders revealed statistical differences for qCSA in favor of responders (group $\times$ set: $P = 0.0035$) for both 1 set (EMD: 2.31 cm^2 , 95% CI: $1.42\text{--}3.20 \text{ cm}^2$) and 4 sets (EMD: 1.12 cm^2 , 95% CI: $0.24\text{--}2.01 \text{ cm}^2$) (all $P < 0.05$). Individual hypertrophic response and differences between 1 and 4 sets for each responsiveness group is depicted in Fig. 2, A and B, respectively.

Maximum Dynamic Muscle Strength (1-RM)

Nonresponders increased the 1-RM pre- to postintervention for 1 set (time $\times$ set: $P = 0.007$; EMD: 2.98 kg ; 95% CI: $0.88\text{--}5.08 \text{ kg}$, $P = 0.0018$) and 4 sets (EMD: 6.11 kg , 95% CI: $4.01\text{--}8.21 \text{ kg}$, $P < 0.0001$). The 1-RM was statistically higher

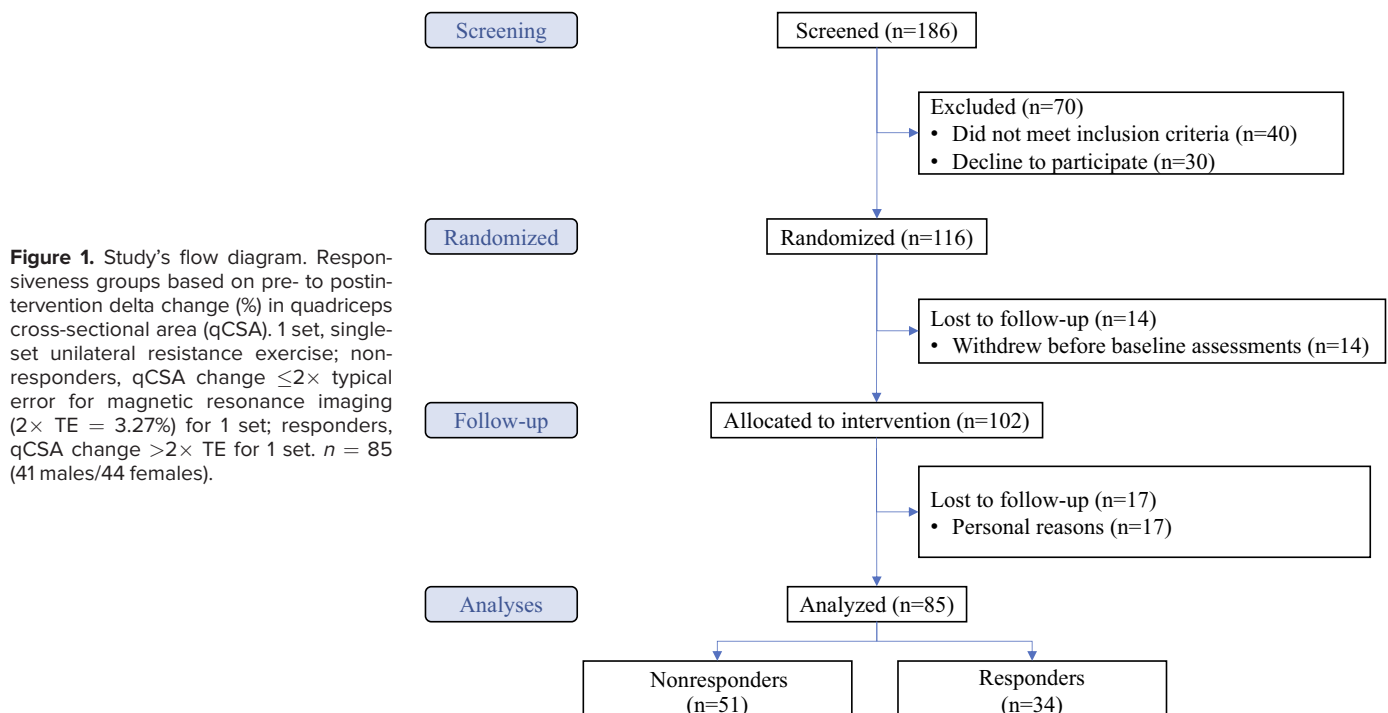


Table 2. Pre- to postintervention values for quadriceps cross-sectional area and maximum dynamic muscle strength (1-RM)

Groups	Quadriceps Cross-Sectional Area, cm ²			
	1 Set		4 Sets	
	Pre	Post	Pre	Post
Nonresponders	54.7±13.4	55.2±13.5	54.3±12.4	57.3±13.3* [#]
Responders [#]	51.4±13.1	54.2±13.4	52.2±13.4	56.4±14.4
Groups	Maximum Dynamic Muscle Strength, kg			
	1 Set		4 Sets	
	Pre	Post	Pre	Post
Nonresponders	45.7±20.2	48.6±21.9*	46.4±18.8	52.5±21.9* [#]
Responders [#]	44.5±19.1	48.3±21	44.6±18.3	50.2±20

Data are presented as means ± standard deviation. Responsiveness groups based on pre- to postintervention change (%) for quadriceps cross-sectional area (qCSA) to 1 set. 1 set, single-set unilateral resistance exercise; nonresponders: qCSA change $\leq 2 \times$ typical error for magnetic resonance imaging ($2 \times \text{TE} = 3.27\%$) for 1 set; responders, qCSA change $> 2 \times \text{TE}$ for 1 set. *Different from preintervention values for the same set ($P < 0.05$); [#]different from postintervention values for 1 set; [#]main time effect ($P < 0.05$). 1-RM, one-repetition maximum.

at postintervention for the 4 sets protocol compared with 1 set (EMD: 3.9 kg; 95% CI: 1.8–6.0, $P < 0.0001$) (Table 2). In contrast, responders improved 1-RM following RT intervention (time: $P < 0.001$; EMD: 5.42 kg, 95% CI: 3.93–6.9 kg), but with no statistical difference between sets (time $\times$ set: $P = 0.23$; 1 set EMD: 4.54 kg, 95% CI: 1.79–7.29 kg; 4 sets EMD:

6.29 kg, 95% CI: 3.54–9.04 kg; 1 set vs. 4 sets postintervention EMD: 1.84 kg, 95% CI: –0.92 to 4.61 kg) (Table 2). Comparison between nonresponders versus responders revealed statistically significant group versus set interaction for 1-RM ($P = 0.03$), but no statistical difference at the post hoc level for neither 1 set (EMD: 1.76 kg, 95% CI: –1.12 to 4.66 kg) nor 4 sets (EMD: 0.007 kg, 95% CI: –2.89 to 2.90 kg) (all $P > 0.05$). Individual changes in 1-RM and differences between 1 and 4 sets for each responsiveness group is depicted in Fig. 2, C and D, respectively.

Total Training Volume

As per design, total training volume was higher for the higher RT volume protocol when compared with the lower one (EMD: 13,097 kg, 95% CI: 11,801–14,394 kg, $P < 0.001$). We did not observe statistical differences between nonresponders and responders for total training volume ($P < 0.05$).

DISCUSSION

Using a within-subject unilateral RT design, we aimed to investigate the effects of RT volume manipulation on whole muscle tissue hypertrophic adaptations among nonresponders and responders to a low RT dose. In addition, we investigated the effects of RT volume manipulation on muscle

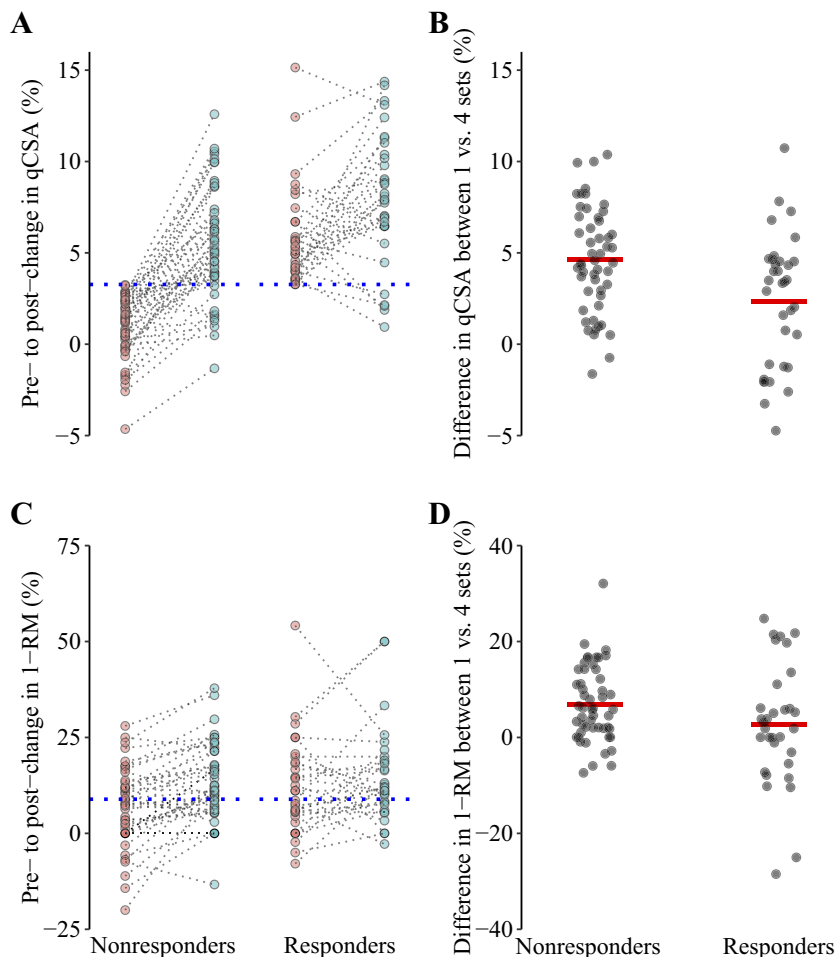


Figure 2. Individual pre- to postintervention delta change (%) in quadriceps muscle cross-sectional area (qCSA) and maximum dynamic strength [1-repetition maximum (RM)] for nonresponders and responders (A and C, respectively). B and D: delta differences in qCSA and 1-RM between 1 vs. 4 sets, respectively. 1 set, single-set unilateral resistance exercise (red); 4 sets, four-set unilateral resistance exercise (green). Nonresponders, qCSA change $\leq 2 \times$ typical error for magnetic resonance imaging ($2 \times \text{TE} = 3.27\%$) for 1 set; responders, qCSA change $> 2 \times \text{TE}$ for 1 set. Dotted blue line: $2 \times \text{TE}$ for magnetic resonance imaging (3.27%) and 1-RM (8.90%). Values are presented as individual data (dots) and mean (red line). $n = 85$ (41 males/44 females).

strength in these responsiveness groups. Our primary results demonstrated that increases in RT volume mitigated nonresponsiveness by improving whole muscle tissue hypertrophy as well as strength gains among nonresponders. Responders improved qCSA, with a tendency toward higher benefit from the 4 sets RT protocol; on the other hand, 1-RM increased similarly irrespectively of RT volume prescription.

It has been hypothesized that nonresponsiveness may be linked with an inadequate training stimulus, which suggests that changes/adjustments in exercise dose could mitigate this phenomenon (12, 13). In fact, studies involving aerobic-based training interventions have demonstrated that higher exercise dose is associated with a lower incidence of nonresponders (15). Also, increases in exercise dose (i.e., increased weekly training frequency) promoted gains in cardiorespiratory fitness in individuals initially classified as nonresponders (14). It has also been suggested that distinct RT protocols could also reduce the rate of nonresponsiveness following intervention (29). Herein, we demonstrated that a higher RT volume (i.e., 4 sets) counteracted nonresponsiveness by increasing not only whole muscle tissue hypertrophy, but also strength in those not responding to a lower RT volume (i.e., 1 set). To the best of our knowledge, only one other study to date addressed this topic in older adults, with contrasting results (30). The authors showed that muscle hypertrophy nonresponsiveness was not altered by increasing training volume and duration. Although hard to reconcile, discrepant results could be related to differences in experimental design (e.g., progressive increases in RT volume vs. RT volume manipulation within the same period) and methods for assessing muscle hypertrophy [e.g., dual-energy X-ray absorptiometry vs. MRI, with the first being less sensible to assess changes in muscle size compared with the gold-standard method used herein (41)]. Our results, on the other hand, corroborate with Hammarström et al. (42) showing a responsiveness effect of volume in a cohort of younger individuals with an overall higher RT volume protocol (~6 vs. 18 weekly sets). Altogether, our data suggest that RT volume can be a valuable and effective strategy to improve muscle mass and strength among nonresponder older adults, thereby mitigating age-related morphological and functional declines seen with aging.

Despite the overall beneficial effect of RT volume against nonresponsiveness, our intraindividual data analysis revealed that 10 nonresponders (out of 51, ~19%) did not increase muscle size when exercising at higher RT volume (Fig. 2A: values for both sets are $<2 \times$ TE). This is in line with findings from Nunes et al. (30). Moreover, a recent study suggested intrinsic biological factors (e.g., satellite cells and myonuclei content) possibly exert a more pronounced effect on muscle hypertrophic adaptations than manipulations of RT variables per se (21). In this context, those who did not respond to 4 sets may be resistant to the anabolic effects of RT due to a partial inability to activate key signaling pathways to support muscle growth (10, 12, 43). Nonetheless, it is still plausible that these individuals could have still benefited from higher RT volumes, longer training intervention, combining training modalities (e.g., aerobic and RT), lifestyle behavior changes, and/or additional nutritional or adjuvant strategies (13, 44–47), warranting further investigations on the topic.

Also, one may argue that the number of nonresponders observed in the present study (i.e., 60%) is higher than that of other RT protocols (ranging from ~2% to ~50%) (5–9, 11, 23–25, 30, 48, 49). Although we cannot not rule out that the slightly higher prevalence of nonresponders in our sample may be related to the low RT volume, previous data indicate an inverted U-shaped RT dose-response curve, with both “too low” and “too high” RT doses being linked with a higher nonresponder rate (11), suggesting an individual-dependent phenomenon that hampers the determination of an optimum RT dose.

This notion gains traction by looking at our intraindividual data showing that while most responders (17, ~51%) experienced additional gains in whole muscle tissue size following higher RT volume (Fig. 2A), which is in line with our grouped-based analysis (Table 2) and previous reports (25, 30, 42), some individuals (5, ~15%) had negative effects from exercising with higher RT volume (Fig. 2A), supporting previous findings suggesting that elevated levels of stress/effort can blunt hypertrophic outcomes (11, 25). Taken together, we speculate that some individuals may require fine adjustments on training workload (e.g., longer recovery periods), suggesting that an RT prescription that considers individual biological characteristics, while avoiding the notion of “one RT dose fits all,” is a promising strategy to optimize muscle hypertrophic response to RT. Finally, it is also worth highlighting that despite the substantial heterogeneity in response magnitude, all individuals classified as responders already benefited from a lower RT dose, indicating that they appear to be highly mechano-responsive to even minimal stimulation, and thus, are more prone to benefit from RT interventions. Future studies should investigate physiological and perception clues that could be used to improve exercise prescription based on individual needs, as well as the underlying biological pathways that differentiate individuals' responses to RT.

Finally, in the present study, we classified individuals as nonresponders or responders based on our primary outcome of interest (i.e., whole muscle tissue size). However, we also included measurements of muscle strength, provided that lower limb muscle strength correlates with muscle function, quality of life, and lifespan in older individuals (50–52). As discussed earlier, average group data indicate an agreement between gains in muscle mass and strength. Intraindividual data analysis, on the other hand, suggests otherwise. For instance, 19 individuals (38%) classified as nonresponders would be considered responders if the criteria used was based on the measurement error ($2 \times$ TE = 8.90%) for the 1-RM measures, whereas 16 responders (48%) would be classified as nonresponders (Supplemental Material 1). These results highlight the notion that responsiveness is outcome specific and likely different depending on the physiological system (adaptations in 1-RM, for instance, are significantly affected by neuromuscular communication efficiency). Future studies should investigate strategies to cluster individual responses based on multiple outcomes of interest, as it is well established that RT has pleiotropic effects (e.g., gains in muscle mass, strength, power, and functionality), particularly for older populations.

This study is not without limitations. We were not able to control for some external factors (e.g., sleep pattern, stress

level, habitual physical activity, and/or sedentary behavior) that may impact the interindividual response variability to the intervention, though this was mitigated by the within-subject design. To mitigate differences in training status as a confounding factor, we only included untrained individuals, which preclude the extrapolation of our findings to more well-trained individuals. The qCSA measure was performed only in the middle of the quadriceps and does not account for differences in hypertrophic response along the length of the whole muscle (53). Also, unilateral RT protocols are prone to cross-education effect (54); nonetheless, it has been suggested that this effect may be minimal when both limbs are assigned to RT intervention (55). Finally, even though we showed an overall beneficial effect of RT volume to mitigate nonresponsiveness, we only tested two doses of RT volume (and for a limited extent of time; i.e., 10 wk). Future studies should further investigate longer training durations in combination with different doses of RT.

Conclusions

Increasing RT volume mitigated nonresponsiveness by improving whole muscle tissue hypertrophy as well as promoting strength gains among nonresponders. RT volume manipulation emerges as a simple and effective potential strategy to improve muscle hypertrophy nonresponsiveness in older adults, and thereby mitigate age-related morphological and functional declines and promote a healthier lifespan. In addition, it could most likely be used to further improve hypertrophic outcomes in responders. Future studies should investigate what are the mechanisms underpinning nonresponsiveness and responsiveness to RT, particularly in light of the wide variability that continues to exist across the spectrum of exercise responsiveness.

DATA AVAILABILITY

Data will be made available upon reasonable request.

SUPPLEMENTAL DATA

Supplemental Material 1: <https://doi.org/10.6084/m9.figshare.24841239>.

GRANTS

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

M.E.L. and H.R. conceived and designed research; M.E.L., M.S.C., G.T. F.C.V., I.L., F.D., and A.Y.A. performed experiments; M.E.L., H.R., and A.E.L. analyzed data; M.E.L., H.R., C.A.L., and B.G. interpreted results of experiments; M.E.L. and H.R. drafted manuscript; M.E.L., M.B., F.C.V., M.S.C., G.T., I.L., F.D., K.M.L., D.J.D., J.S.M., C.M.D., A.E.L., L.A.R.C., A.Y.A., C.A.L., B.G., and H.R. edited and revised

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